Pos	Part no.	Description	Qty	Make/buy	Material / spec	kg each
		A ROTOR				
1	NM8-100	Blade Ø8.0 m, glass/epoxy, carbon UD caps (B1–B3)	3	make	E-glass/epoxy, carbon UD caps, PVC core	34.1
2	NM8-105	Root insert M16 (12 per blade, bonded)	36	make	C45E+QT zinc-flake coated	0.8
3	NM8-200	Hub weldment: shell Ø530 × 10, 3 necks Ø360, drum flange	1	make	S355NL, welded, stress relieved, machined	179.8
4	NM8-210	Spinner (GRP 4 mm) with blade boots	1	make	glass/polyester, gelcoat RAL 9010	27.6
5	NM8-250	Pitch bearings Dpw 300 four-point (3, bought)	1	buy	bought: 42CrMo4 rings	76.2
6	NM8-260	Pitch cranks r140 with pins (3)	1	make	42CrMo4+QT / S355 tube	30.0
7	NM8-261	Pitch links 220 mm with spherical eyes (3)	1	make	S355 / bought rod ends	4.5
8	NM8-262	Yoke + 3 guide rods	1	make	S355 plate / C45 chromed rods	8.0
9	NM8-270	Electric pitch cylinder ~28.5 kN rated, 204 mm stroke + motor	1	buy	bought	14.0
10	NM8-280	Supercap module + pitch drive/controller (bought)	1	buy	bought	6.5
11	NM8-290	Nose frame (actuator + electronics carrier)	1	make	S355 rings + tube struts	6.0
		B DRIVETRAIN AND GENERATOR REV D				
12	NM8-311	Third caliper bracket (135°)	1	make	S355J2+N 25 plate, welded	7.0
13	NM8-913	Tapered roller bearing 32028 X (front + rear)	2	buy	bought 32028 X	4.0
14	NM8-914	Overspeed sensors M18 inductive (2, bought)	1	buy	bought M18 PNP	0.4
15	NM8-915	Spring-applied brake caliper 4.8 kNm (3 on the disc)	3	buy	bought, spring-applied	9.0
16	NP1-100D	Spindle Rev D (30 mm longer between the rear seat and the flan	1	make	42CrMo4+QT forged	59.2
17	NP1-101	Front spacer / seal counterface Ø180 (Rev B)	1	make	42CrMo4+QT, induction hardened	1.4
18	NP1-102	Rear seal sleeve Ø195 on the Ø155 shoulder (Rev B)	1	make	42CrMo4+QT, induction hardened	1.3
19	NP1-200D	Stator carrier Rev D, 45 dovetail grooves, rim 104 mm	1	make	S355J2+N	45.4
20	NP1-201D	Stator Rev D: 45 tooth-segment stacks, 104 mm (7 magnet rows)	1	make	M270-35A 0.35 mm, self-bonding	71.0
21	NP1-202	Stator front clamp ring	1	make	1.4404 (non-magnetic)	2.2
22	NP1-203D	Coils on segments, 45 × 28 turns, 104 mm stack	1	make	Cu enamelled Ø1.55 class 200, NKN 0.30	20.0
23	NP1-210D	Rotor drum weldment Rev D (shell +30; Rev C fins and ribs, see	1	make	S355J2+N	95.5
24	NP1-211-F	Front bearing seal cover	1	make	S355J2	1.5
25	NP1-211-R	Rear bearing seal cover, bore Ø198 (Rev B)	1	make	S355J2	1.4
26	NP1-212C	Drum rear cover with 24 fan blades (Rev C; blades not modelled	1	make	EN AW-5083-H111	7.2
27	NP1-220F	Magnets 280 × N45SH 30×15×12 in 7 rows + comb + sleeve	1	make	NdFeB N45SH	15.0
28	NP1-230	Brake disc Ø880/Ø652 with 12 lock holes (Rev B)	1	make	C45E+N 16 mm	33.6
29	NP1-310	Rotor brake caliper bracket (left, Rev B)	1	make	S355J2+N 25 plate	18.1
30	NP1-320	Rotor lock pin Ø20 (engaged, top)	1	make	42CrMo4+QT	0.5
31	NP1-321	Lock pin holder with 2 overspeed sensor seats	1	make	S355J2	24.5
		C NACELLE				
32	NM8-700-1	Nacelle cover, lower tray (GRP 5 mm)	1	make	glass/polyester, gelcoat RAL 9010	15.7
33	NM8-700-2	Nacelle cover, upper lid (GRP 5 mm)	1	make	glass/polyester, gelcoat RAL 9010	17.7
34	NM8-710	Cover support arms + rear rail	1	make	S235 galv.	8.3
35	NM8-720	Met mast: anemometer, vane, air terminal	1	make	1.4404 tube 42.4 × 2 + bought sensors	4.5
36	NP8-300	Bedplate weldment 12 kW NP8 (plates 20 / 30 / 10)	1	make	S355J2+N plates, stress relieved, machined	124.3
		D YAW SYSTEM				
37	NM8-911	Yaw slewing ring Dpw 410, z165 m4	1	buy	bought: 42CrMo4, induction hardened teeth	66.0
38	NM8-912	Yaw gearmotor 0.25 kW, output ≥ 2.3 kNm static, motor brake	2	buy	bought	19.0
39	NP1-420	Yaw drive bracket	2	make	S355J2+N	7.7
40	NY-400	Tower-top adapter with friction track	1	make	S355J2+N weldment	118.3
41	NY-410	Friction pad unit	2	make	S355 / DIN 2093 springs / organic pads	14.0
42	NY-430	Yaw lock (screw block)	1	make	42CrMo4 / S355	6.5
43	NY-440	Yaw encoder + cam limit switch unit	1	make	S355 + absolute encoder + cam switch	6.0
44	NY-460	Cable hanger tube + loop cleat	1	make	S355 tube 168.3 × 4.5	10.0
		E ELECTRICAL				
45	NM8-600	Braking / dump resistor box (vented)	1	make	1.4301 box + steel-grid resistors	18.0
46	NM8-620	Tower base cabinet IP55 800 × 600 × 300 on stand	1	make	stand S235 galv.; bought enclosure	95.0
47	NM8-730	Nacelle junction box IP66 (400 × 340 × 150)	1	make	bought: 1.4301 enclosure	9.0

NOTES

1. Masses from the CAD model (cad/turbine_m8/model8.py) for made parts; supplier data or the doc 35 budget for bought and pitch parts. Full BOM: bom8.csv.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

BILL OF MATERIALS (1/2)

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-002	A · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see lines	see lines	Claude / not checked